



Explore the exponential growth of renewable energy based on IRENA Global Energy Statistics.
Witness the transformation of our energy system toward sustainability.

4,351.1

Total Renewable Capacity (GW)
Global renewable capacity 2024

3,500%

Solar Growth Since 2010
Fastest growing technology

\$1.8T

Annual Investment (2023)
Record investment levels

30.1%

Share of Global Electricity
Up from 20% in 2010

 Data Controls

Technology

 All Technologies

 Solar

 Wind

 Hydro

 Geothermal

 Biomass

Time Range

 All Time (2000-2024)

 Last Decade (2014-2024)

Current Selection:

All Technologies
25 years of data



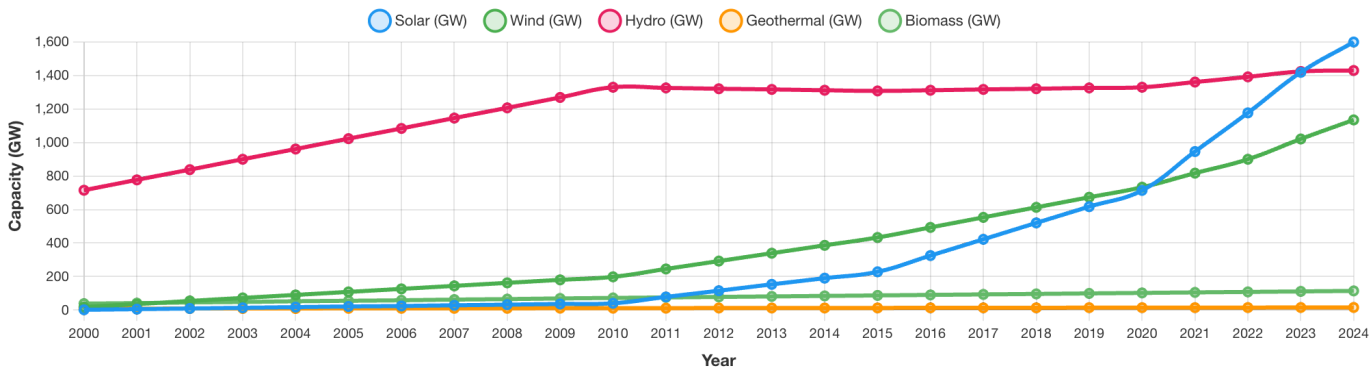
Sustainability Chatbot

Renewable Energy Capacity Growth

Data Source: International Renewable Energy Agency (IRENA) Global Energy Statistics
Time Range: 2000-2024 (25 years)
Technology: All renewable technologies

🌟 **2024 BREAKTHROUGH:** Solar energy surpassed hydroelectric power in global capacity, marking a historic shift in the renewable energy landscape!

⚠️ **POLICY UNCERTAINTY:** While solar has achieved historic growth under supportive policies, potential future policy changes could impact growth rates. The Inflation Reduction Act (IRA) has been crucial for recent acceleration.

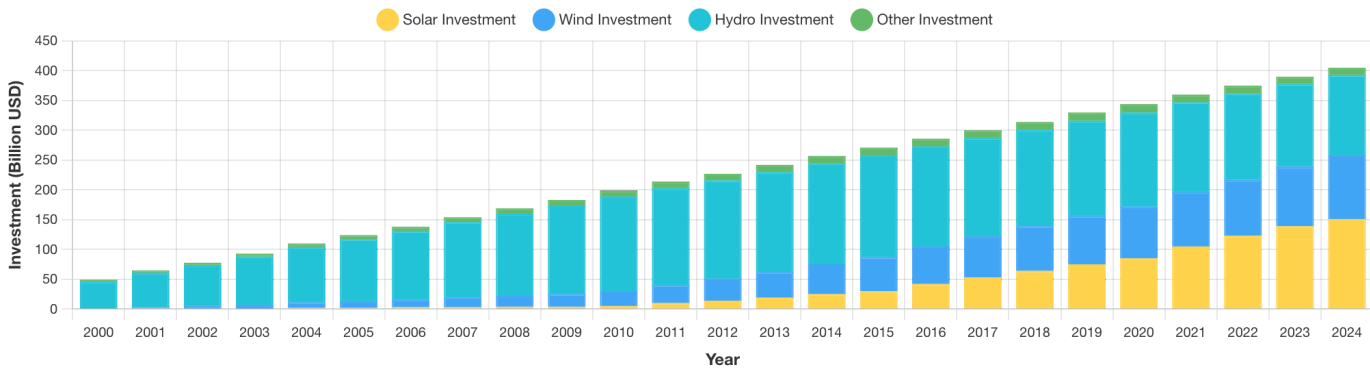


🔍 **Key Finding**

Solar energy made history in 2024 by overtaking hydroelectric power in global capacity. Combined with wind, these variable renewables now dominate new capacity additions, but face policy uncertainty.

Global Renewable Energy Investment

Data Source: IRENA Global Energy Statistics & Investment Trends
Annual investment in renewable power generation capacity (excludes EVs, batteries, and other clean tech). Total clean energy investment including all technologies reached \$1.8 trillion in 2023.



🌟 **Why Renewable Energy is Our Hope for the Future**

Renewable energy has experienced unprecedented growth over the past decade, transforming from a niche technology to the dominant force in global energy expansion. **Solar capacity has grown 3,500% since 2010**, while wind power has increased by 400%. This explosive growth represents humanity's greatest hope for combating climate change while building a sustainable, prosperous future.

💰 Why Renewable Energy's \$395B Matters in a \$1.8 Trillion Clean Energy World

Our Graph Above Shows: \$395B (Renewable Power Generation Only)

- ☀️ Solar power plants and installations
- 💨 Wind farms (onshore and offshore)
- 💧 Hydroelectric facilities
- 🌋 Geothermal power plants
- 🌱 Biomass energy facilities

Total Clean Energy Investment: \$1.8 Trillion (2023)

Renewable Power: \$395B Solar, wind, hydro facilities	Electric Vehicles: \$270B EVs, charging infrastructure	Grid & Storage: \$310B Smart grids, batteries
Energy Efficiency: \$180B Heat pumps, insulation	Hydrogen & Other: \$220B Green hydrogen, CCUS	Nuclear: \$65B New nuclear plants

🚀 **Why This Matters:**

Total clean energy investment reached \$1.8 trillion in 2024, but renewable power generation received "only" \$395B (22%).

Why does this seemingly small slice matter so much?Because renewables are the foundation that enables everything else - without cheap, clean electricity from solar and wind, electric vehicles, heat pumps, and green hydrogen simply wouldn't be viable.

Renewables aren't just 22% of clean energy investment—they're the catalyst that makes the other 78% economically possible. This is why the renewable energy revolution is the most critical piece of our climate solution.

🔍 Investment Surge

Renewable power generation investment has grown 900% since 2010, reaching \$395 billion in 2023. When including all clean energy technologies (EVs, batteries, heat pumps), total investment reached \$1.8 trillion. Solar and wind are now the cheapest sources of electricity in most countries, driving continued investment growth.

About This Project

Created for the **Viz Con 2025** contest at **Analyticon**(Amazon-wide Conference on Data Engineering, Science and Analytics). This interactive application explores the connections between human activity and environmental conservation through compelling data narratives using real datasets.

Data Sources

- 🌡️ **Temperature Data:**
NASA GISTEMP v4 (Goddard Institute)
- ⚡ **Renewable Energy:**
IRENA Global Energy Statistics
- 🔍 **Critical Minerals:**
IEA Critical Minerals Data Explorer
- 🧊 **Sea Ice Data:**
NSIDC Sea Ice Index
- 🏭 **CO2 Data:**
NOAA Mauna Loa Observatory

Technology & AI

- Frontend:** React.js, D3.js, Chart.js, Framer Motion
- Backend:** Python Flask, NumPy, Pandas
- AI Integration:** Pattern recognition, correlation analysis, predictive modeling, and natural language insight generation
- AI Chatbot:** Amazon Bedrock (Nova Pro Model), API Gateway, AWS Lambda, S3, IAM
- Deployment:** AWS Amplify with automated CI/CD pipeline

Viz Con 2025 Entry

Entry for the **2025 Viz Con contest**, part of the
Analyticon - Amazon-wide Conference on Data
Engineering, Science and Analytics

Challenge: "Visualizing a Sustainable
Tomorrow"

Approach: Multi-dataset integration with real
NASA GISTEMP, NOAA, IRENA, and IEA data

Goal: Inspire climate action through compelling
data storytelling

Features: Interactive visualizations, correlation
analysis, predictive modeling, and accessibility-
first design

Built with  by [@ygal](#) and his best friend Amazon Q 

Powered by  | NASA, IRENA, IEA, NSIDC & NOAA data